Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/11/2013
PI Name:	Weil, Michael Ph.D.		
Project Title:	Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Fort Collins	**State:**	CO
Zip Code:	80521-2807	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	11/10/2011	**End Date:**	11/09/2015
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	No changes in year 2.		
COI Name (Institution):	Belknap, John (Oregon Health & Science University) Raber, Jacob (Oregon Health and Science University) Hall, Eric (Columbia University) Kleiman, Norman (Columbia University)		
Grant/Contract No.:	NNX12AB54G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal has two key objectives. The first is to compare the spectrum of tumor types that arises in a genetically diverse mouse population irradiated with HZE ions to the spectra arising in similar unirradiated or gamma-ray irradiated populations. The second objective is to identify genetic differences between individual outbred mice that make them susceptible or resistant to the induction of specific types of tumors by radiation and then to determine if the same genetic factors that control sensitivity to gamma-ray induced tumors also control sensitivity to HZE ion induced tumors. To accomplish these objectives we will irradiate mice from an outbred, heterogeneous stock, HS/NPT, with HZE ions or gamma-rays and monitor them along with unirradiated control mice for tumor development. Tumors that arise in the mice will be classified by histological examination and the tumor spectra will be compared between the three groups. We will identify the genetic differences, known as QTL or quantitative trait loci, underlying susceptibility to HZE ion-induced and gamma-ray-induced tumorigenesis and determine their overlap by analyzing single nucleotide polymorphisms (SNP) throughout the genomes of tumor bearing and unaffected mice for association with specific tumor types. The proposed work addresses the solicitation's emphasis on studies of the individual basis of space radiation risk assessment including the "use of genetically diverse populations of mice to represent human heterogeneity that will elucidate the spectrum of tumor types caused by HZE nuclei as well as possible differences compared to gamma-rays." In addition, it addresses NASA identified knowledge gaps related to uncertainties in risk projections arising from individual radiation sensitivity and gender. Finally, the proposed study has the potential to identify experimental models for tumor development in tissues for which such models are lacking or limited such as colon, stomach, bladder, skin, esophagus, and brain.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Radiation carcinogenesis is a matter of concern for individuals exposed to radiation for medical diagnosis or treatment, for occupational reasons, or because of accident. This study will provide basic information about genetic susceptibility to radiation-induced cancers.
Task Progress:	Year 2 Task Progress Background. In year 1 1800 HS/Npt mice were bred at Colorado State University (CSU) and irradiated or sham irradiated at NSRL or the Medical Department at the Brookhaven National Laboratory and returned to CSU for monitoring. These mice have been phenotyped for neurobehavioral endpoints and are being followed for tumor development and ocular damage. Year 2: Tumorigenesis. Over the past year the mice have been monitored daily for morbidity. About 500 mice that became moribund or showed signs of cancer were euthanized and necropsied. Tissues were collected from all of these mice and have been prepared for histopathological examination. Histopathology has been completed on about 50 cases. SNP Genotyping. We prepared genomic DNA samples from tail snips from 600 mice and had them genotyped by GeneSeek (Lincoln, NE) for 77,800 SNP markers on the Mega Mouse Universal Genotyping Array (MegaMUGA) platform. Good genotyping data was obtained. Neurobehavioral effects. Drs. Raber, Belknap, and Iancu began a quantitative trait loci (QTL) analysis of a group of mice cognitively tested in Yr 1 of this project. Based on the fear conditioning data, they selected 600 mice for SNP genotyping. To assure quality of the genotyping data, it was done in batches. The first batch of 98 samples was analyzed and verified to be of high quality by John Belknap at OHSU. All 600 samples were genotyped for ~ 77.8K SNP markers and the data will be used to identify loci that control susceptibilities for cognitive effects (contextual and cued fear conditioning) and other outcome measures. Overlaps in susceptibility loci between radiation qualities would suggest overlaps in pathways. The mapping resolution is in the 200 kb to 2 Mb range for this HS mapping population and all of the founder strains have been completely sequenced, so promising candidate genes are very likely to emerge from the study. The remaining 1,200 mice will be genotyped during the last two years of this project. Once we have the genotyping data on these 1,200 mice, we will redo the analysis including all 1,800 mice. Ocular effects. Ocular effects (measurement of radiation cataract onset and progression) were monitored at 4-6 week intervals by veterinary ophthalmology residents trained by Dr. Norman Kleiman of Columbia University. Posterior lens changes, including opacities characteristic of radiation exposure, were scored by a modified Merriam-Focht methodology. While individual mouse genotypes were hidden from the observers, it is clear that the prevalence and severity of posterior subcapsular opacities in irradiated animals is increasing as compared to unexposed controls, which, in general have very mild to no posterior lens changes. In addition to frank opacities, a number of other ocular disorders were noted and recorded, including corneal haze, inflammation and, in a few cases, orbital or intraocular tumors. We hope to better characterize the nature of the eye tumors once the histopathology reports are completed and determine whether there are any significant associations with genotype. Over the past six months, a Virtual Optomotry Testing (VOT) system was installed and a manufacturer’s representative accompanied Dr. Kleiman to CSU for setup and training. A veterinary medical student, with previous experience in VOT during her undergraduate research experience, was recruited to assist in the examinations on a bimonthly basis.
Bibliography Type:	Description: (Last Updated: 04/27/2026)
Abstracts for Journals and Proceedings	Raber J, Belknap J, Kleiman N, Hall E, Ray A, Fallgren C, Kamstock D, Edmondson E, Weil M. "Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice." HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Abstracts for Journals and Proceedings	Raber J, Belknap J, Kleiman N, Hall E, Ray A, Fallgren C, Weil M. "Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012

Abstracts for Journals and Proceedings	Raber J, Turker MS, Stewart B, Eiwaz M, Turner J, Kronenberg A, Belknap J, Weil MM. "Effect of Space Irradiation on cognition in B6D2 and HS/Npt outbred mice." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Weil M. "Characterization of the Tumor Spectrum Arising in HZE Ion Irradiated Outbred Mice." International Symposium on Radiation Science, Taipei, Taiwan, May 20-21, 2013. International Symposium on Radiation Science, Taipei, Taiwan, May 20-21, 2013. , May-2013